5.2. **Knots.**

Definition 5.6 (Knot)**.** A *knot* is a tuple

$$K = (\beta^\star, \rho^\star, \chi^\star, \eta^\star)$$

where $\beta^\star$ is a selected replayable braid, $\rho^\star$ is its terminal state root, $\chi^\star$ is an aggregate execution trace commitment, and $\eta^\star$ is an optional Bitcoin anchor commitment.

Definition 5.7 (Selection functional)**.** A *knot selection functional* is a map

$$\mathcal{K} : \mathfrak{B} \to \mathcal{P}(\mathfrak{B})$$

which, given a finite set $\mathfrak{B}$ of candidate braids, returns the subset of admissible braid winners under local rules.

Definition 5.8 (Heaviest replayable braid rule)**.** A selection rule is *heaviest replayable* if it returns

$$\mathcal{K}(\mathfrak{B}) = \arg\max_{\beta \in \mathfrak{B}_{\mathrm{rep}}} W(\beta)$$

where $\mathfrak{B}_{\mathrm{rep}}$ is the subset of replayable braids.

Proposition 5.9 (Existence of maximizer)**.** *If $\mathfrak{B}_{\mathrm{rep}}$ is finite and nonempty, and $W(\beta)$ is real-valued, then at least one heaviest replayable braid exists.*

Proof. A real-valued function on a finite nonempty set attains a maximum. □

Proposition 5.10 (Selection stability under dominated extensions)**.** *Let $\beta^\star$ be uniquely heaviest replayable in $\mathfrak{B}$. If a new candidate braid $\tilde{\beta}$ enters with*

$$W(\tilde{\beta}) < W(\beta^\star),$$

then $\beta^\star$ remains the unique winner.

Proof. Immediate from the uniqueness of the maximum. □

6. Compression Into Bitcoin Settlement

6.1. **Anchors.**

Definition 6.1 (Anchor commitment)**.** An anchor commitment is a function

$$\eta : \mathcal{R} \times \mathcal{T} \to \mathcal{B}$$

that maps a state root and trace commitment into a Bitcoin-valid settlement artifact.

Definition 6.2 (Compression ratio)**.** If a braid β contains n messages with total serialized payload

$$L(\beta) = \sum_{m \in V_\beta} \ell(m),$$

and its anchor artifact has Bitcoin weight $W_{\mathrm{BTC}}(\eta^\star)$, then define the anchoring compression ratio

$$\Gamma(\beta) = \frac{L(\beta)}{W_{\mathrm{BTC}}(\eta^\star)}.$$